Gel permeation chromatographie (GPC)

Protocol for the laboratory course in Polymer chemistry

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1 Theory

With GPC the molecular weight distribution of a polymer can be determined. The seperation process is basesd on the hydrodynamic volume of a polymer. The polymer solution goes through a column made out of Polystyrene crosslinked with Divinylbenze (stationary phase). The solvent going through the column (mobile phase) is the same solvent that the polymer is dissolved in. In figure 1 a schematic representation of the colums is shown.

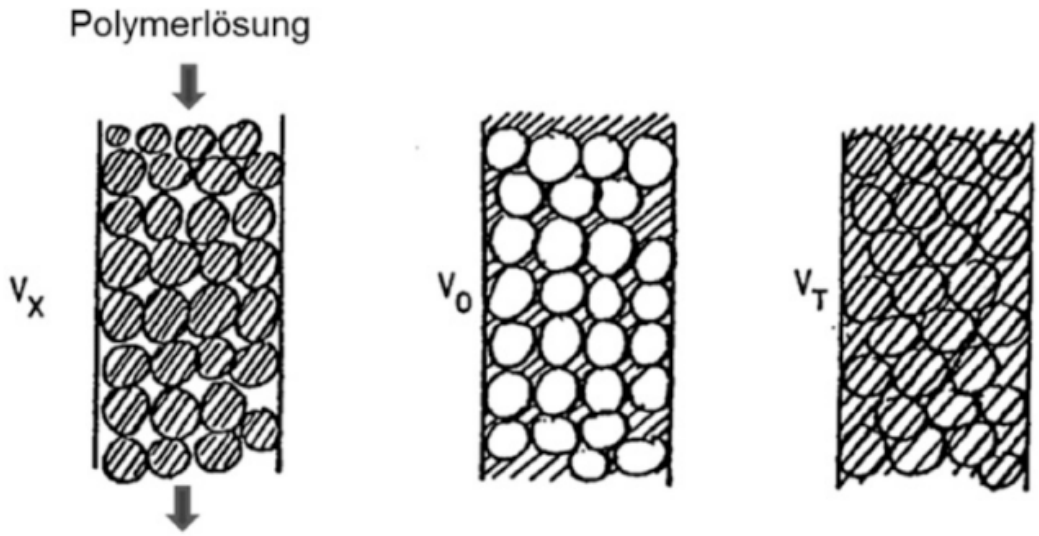

Figure 1: The schematic representation of the column.[1]

V_x is the volume of the stationary phase, V_0 the volume of the mobile phase und V_T the total volume of the column. With these parameters the elution volume can be determinde. This is shown in eqution (1.1).

$$V_\mathrm{e} = V_0 + K_\mathrm{d} \cdot V_\mathrm{x} \tag{1.1}$$

K_d stands for the seperation coefficient, which indicates the fraction of the pore volume. Polymers that are too big dont interact with the column and went straight through it. Polymers that are too small are entering the stationary phase and leaving it quickly after. Polymers that have the right size are entering the stationary phase and stay there for a longer time. K_d equals zero when the the polymer can not enter the stationary phase and equals one when the molecules can enter the stationary phase. With the help of a polymer standard a calibration curve in the form of $\log M = A - B \cdot V_\mathrm{e}$ can be represented. An experimentally determined relationship of $V_\mathrm{e} - \log M$ with a known molecular weight is shown in figure 2.

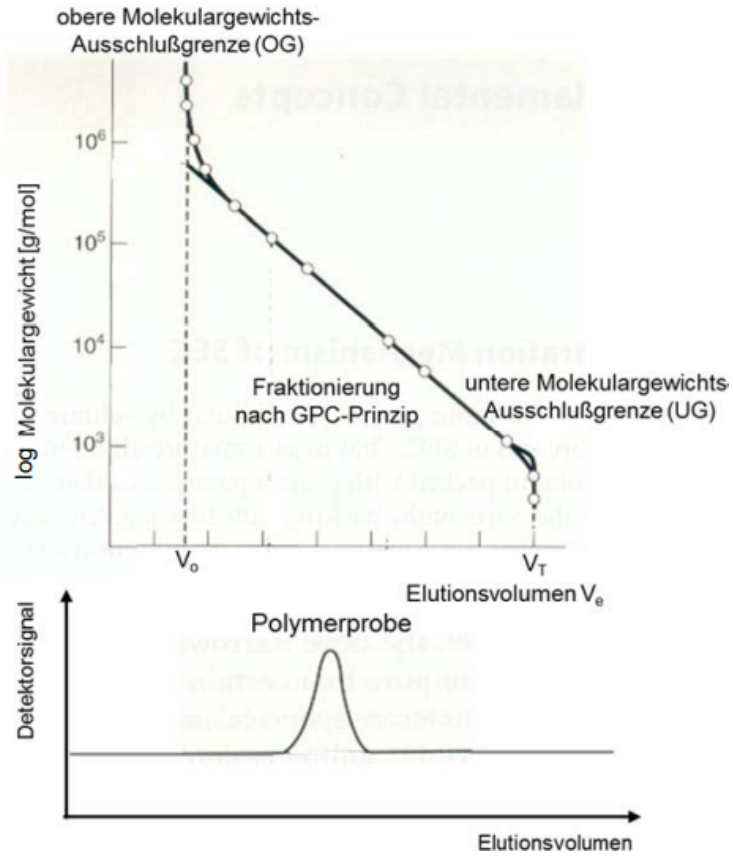

Figure 2: The relation of $V_\mathrm{e} - \log M$ with a known molecular weight.[1]

Another important aspect about GPC are the number of theoretical plates. These are an indicator of the separation performance of a GPC column. With the number of theoretical plates N the „height equivalent theoretical plates“, HETP can bes determined with equation (1.2)[1].

$$\mathrm{HETP} = \frac{L}{N} \tag{1.2}$$

L equals the length of the column.

2 Execution

A calibration was made, using Polystyrene as a standard. The temperature in the GPC were 30 °C and for the standard a 50 µl Polystyrene solution is used. The experiment data for the Analysis was given by the assitant. In table 1 the conditions in the instrumental setup are described.

Table 1: The conditions of the instrumental setup.

Eluent	THF
Injection volume	50 µl
Flow rate	1 mL/min
Temperature	30 °C
Detectors	RI and UV
Columns	SDV
Runtime	30 min (for each sample)

3 Analysis

3.1 Calibration and HETP

The obtained calibration curve is shown in figure 3.

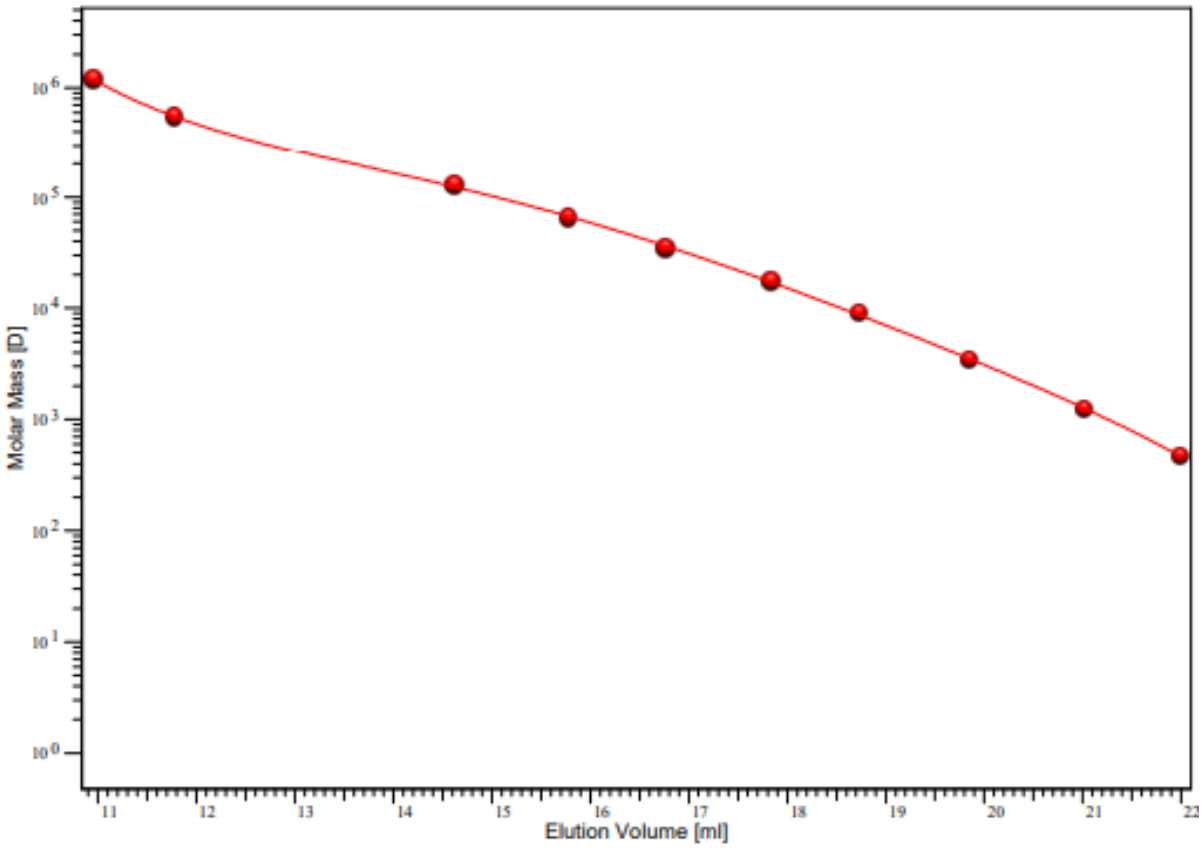

Figure 3: The obtianed calibration curve.

For that the peak maxima of V_e was designated to a defined molar mass. The molar mass of the Polystyrene standard was given by the manufacturer, and V_e measured by the GPC. The fit-type is fifth polynom and equals a deviation of $R = 0,99993$.
With the length of the column and the number of theoretical plates the height equivalent of the theoretical plates HETP can be determined with equation (1.2).

$$\text{HETP} = \frac{0.65\,\text{m}}{33000}$$
$$\text{HETP} = 1.97 \times 10^{-5}\,\text{m}$$

3.2 Polystyrene

In figure 4 the elugram of Polystyrene is shown.

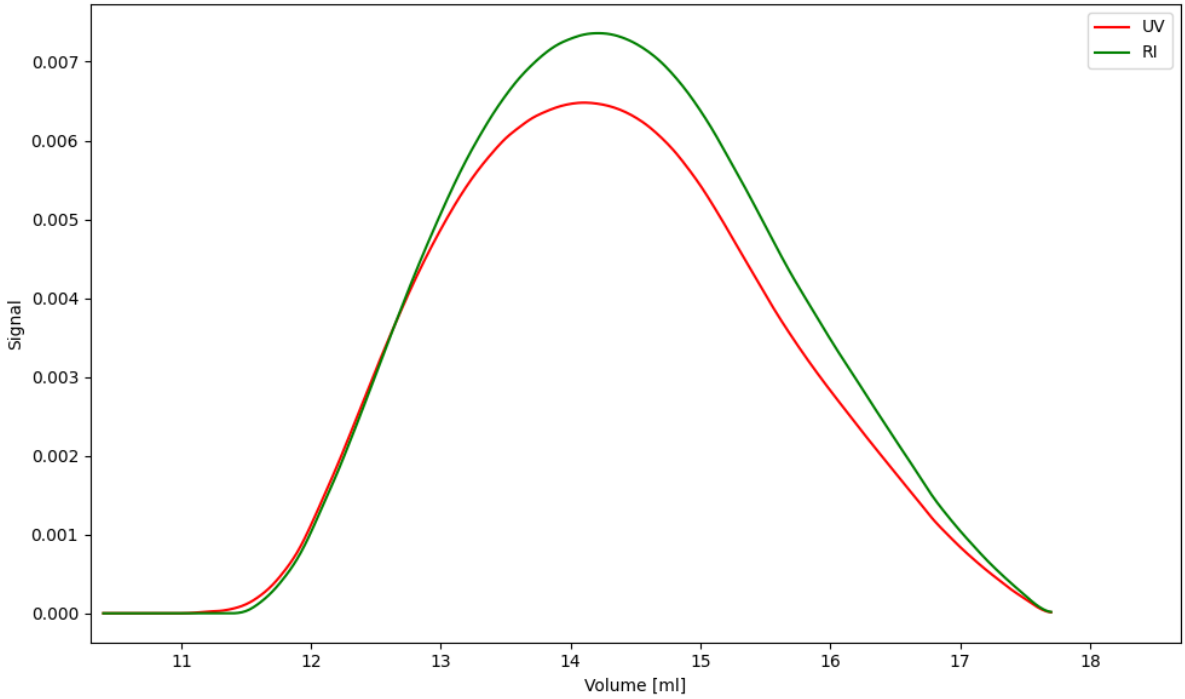

Figure 4: The elugram of Polystyrene. In green, the curve from the RI-detector and in red, the curve of the UV-detector.

Figure 5 shows the molecular weight distribution of Polystyrene.

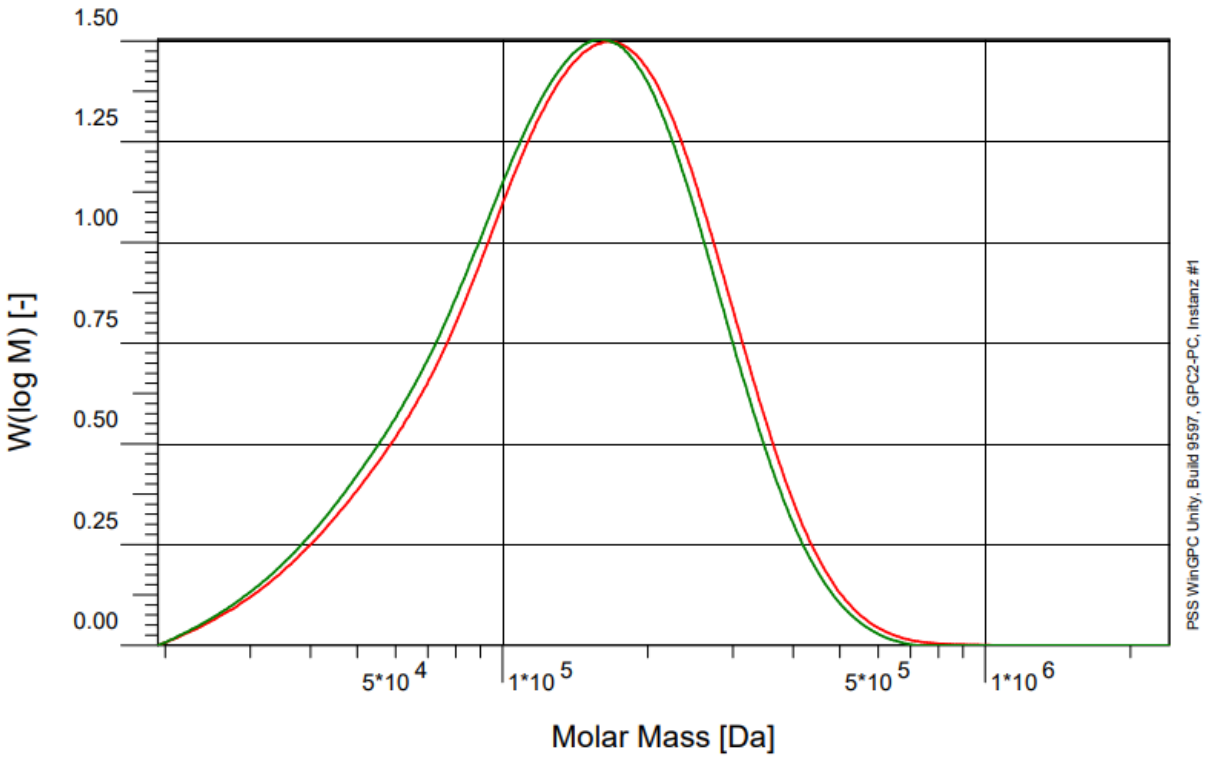

Figure 5: The molecular weight of Polystyrene. In green, the curve from the RI-detector and in red, the curve of the UV-detector.

As can be seen, there is a broad molecular weight distribution in both curves. There is also one peak in the diagram above, which means there aren't any other polymers in the Polystyrene sample. The molecular weights and corresponding PDIs are shown in table 2.

Table 2: The molecular weights and PDIs of Polystyrene.

	UV	RI
$\bar{M}_{\mathrm{n}}$ [g/mol]	$1.154 \cdot 10^5$	$1.108 \cdot 10^5$
$\bar{M}_{\mathrm{w}}$ [g/mol]	$1.693 \cdot 10^5$	$1.616 \cdot 10^5$
PDI	1.468	1.459

There is a difference about 0.61 % in between these two PDIs. Also the PDI is greater than one, which means that not all of the polymers have the same chain length. The theoretical molecular weight from Sigma Aldrich is $M_{\mathrm{w}} = 1.92 \times 10^5$ g/mol. Comparing this with the values from table 2 it can be seen that the theoretical molecular weight is higher than the measured ones. This is because, in theory, it is assumed that each monomer molecule starts its polymerization at the same time, which is not the case in reality.

3.3 Polystyrene-b-Polyisoprene

In figure 6 the elugram of Polystyrene-b-Polyisoprene, synthesized by living-anionic-polymerisation is shown.

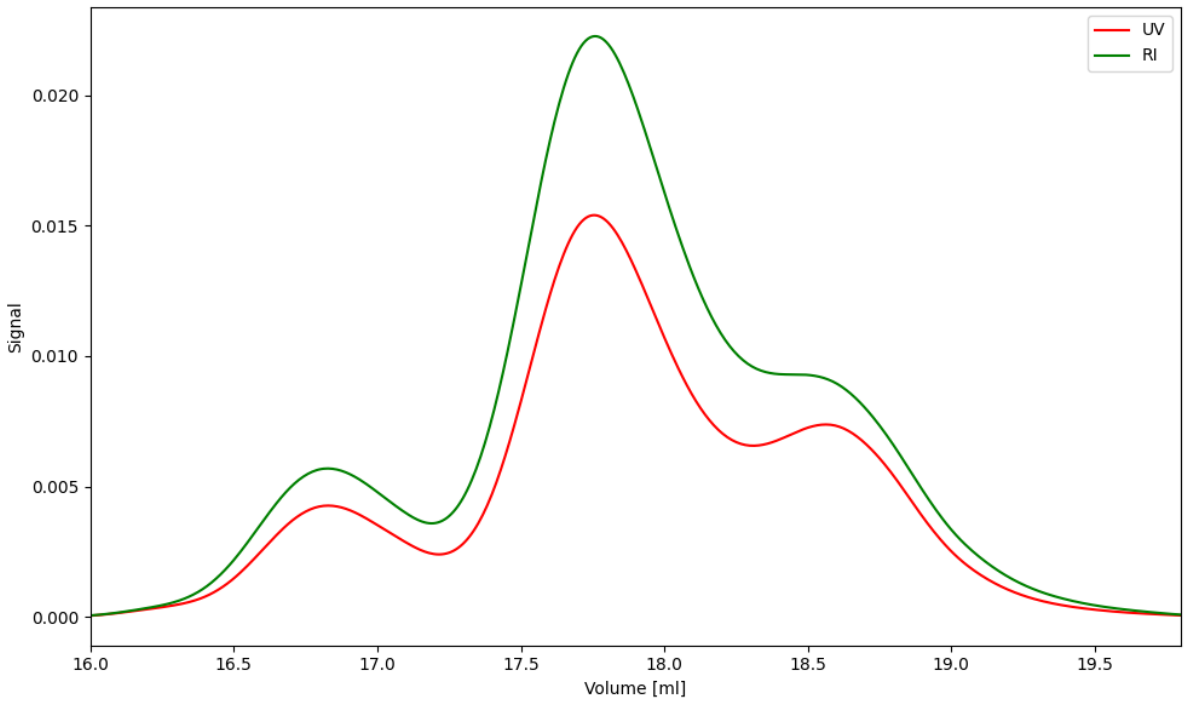

Figure 6: The elugram of Polystyrene. In green, the curve from the RI-detector and in red, the curve of the UV-detector.

In figure 7 the molecular weight distribution of Polystyrene-b-Polyisoprene is shown.

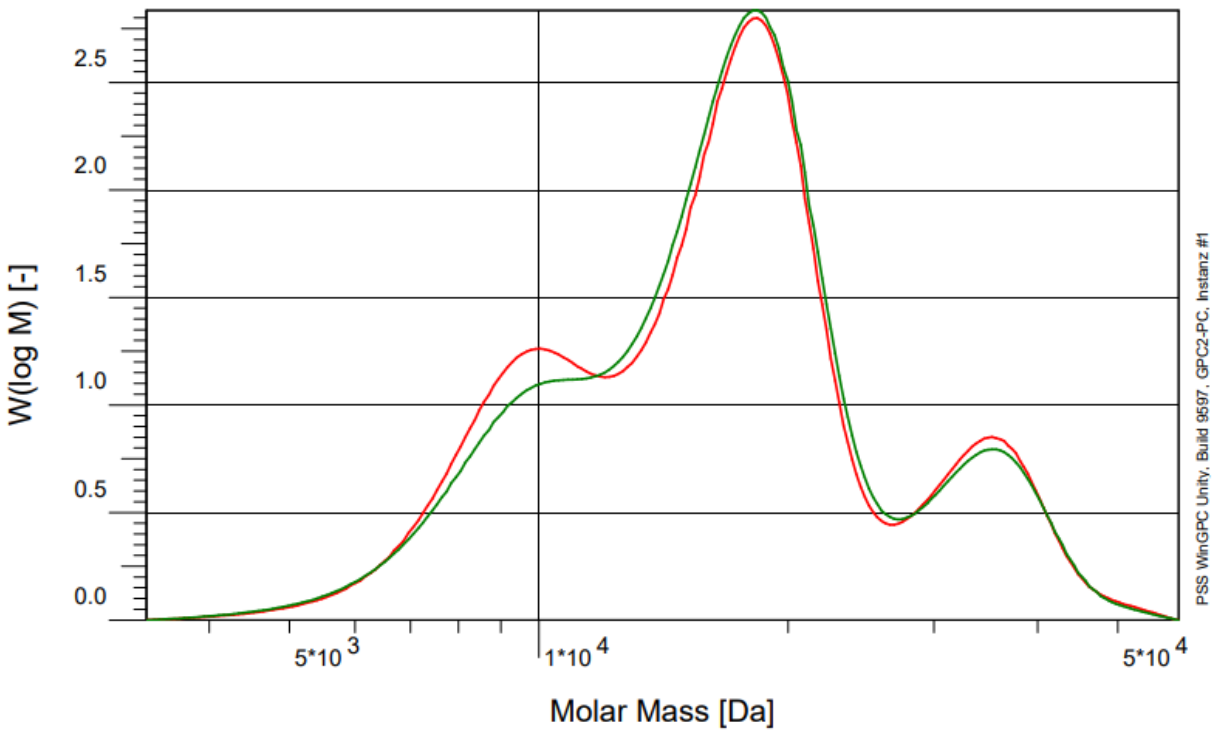

Figure 7: The molecular weight distribution of Polystyrene-b-Polyisoprene. In green, the curve from the RI-detector and in red, the curve of the UV-detector.

Usually GPC gives a Gaussian elution curve. With that a certain molecular weight corresponds to V_e. Synthetic polymers have a narrow or broad distribution, also called polydisperse, while natural polymers are monodispers, which corresponds in a narrow distribution. Another import thing is that smaller Molecules solve faster than bigger molecules. The curve from figure 4 for example is a Gaussian elution curve with one signal, while the elution curve from figure 6 has three. The signal with the highest intensity corresponds to the desired polymer. The signal with lower molar mass originates from polystyrene that was terminated upon the addition of isoprene and the signal with higher molar mass is from the dimer formed in the presence of oxygen. The molecular weights and corresponding PDIs are shown in table 3.

Table 3: The molecular weights and PDIs of Polystyrene-b-Polyisoprene.

	UV	RI
$\bar{M}_n$ [g/mol]	$1.458 \cdot 10^4$	$1.481 \cdot 10^4$
$\bar{M}_w$ [g/mol]	$1.813 \cdot 10^4$	$1.825 \cdot 10^4$
PDI	1.243	1.233

There is a difference about 0.80 % in between these two PDIs. Also the PDI is greater than one, which means that not all of the polymers have the same chain length. By comparing the PDIs from table 2 and 3, the PDI from the living anionic polymerisation is smaller than the PDI of the Polystyrene. This means that there where longer polymer chains sythesized via living anionic polymerisation.

4 Questions

1. What components make up the column volume?

- V_T is the total volume of the column, V_x is the volume of the stationary phase and V_0 is the volume of the mobile phase.

$$V_\mathrm{T} = V_\mathrm{x} + V_0$$

2. What absolute methods for determining molar mass do you know, and how can you determine molecular weight distributions?

-For example membrane osmomeometry is a absolute methode based on the colligative propertys and osomsotic pressure. With membrane osmomeometry $\bar{M}_\mathrm{n}$ can be determined. $\bar{M}_\mathrm{w}$ can be determined with light scattering- With thet the PDI can be calculated PDI $= \bar{M}_\mathrm{w}/\bar{M}_\mathrm{n}$. The moluclar weight distribution can also be determined via GPC.

3. How can solvents be degassed continuously?

- With the help of a high pressure pump, pumping solvent continuousely into the system.

4. Which parameters influence the separation performance in gel chromatography?

- The size of the stationary phase, the solvent, the pressure, the temperature, the flow rate and K_d.

5. What requirements must be met by the pump and detectors?

- The pump has to be consistent, so there is a constant flow rate. Also it needs to be a high pressure pump. The detectors have to be able to detect the polymers.

6. How does a differential refractometer work?

- The difference between the refractive index of mobile phase and polymer sample is measured. By this the conecentration of a dissolved polymer can be determined. It is also important to keep the measurement on a constant temperature.

5 References

[1] *"Praktikumsskript deutsch März 2025 "*, Institut für Polymerchemie, pp. 133–142.